

Mohamed Ismail Mohamed Tawfik

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Education

Faculty of Computers and Artificial Intelligence – Cairo University

Information Systems Department (2023-2027)

Graduation year : 2027

GPA: 3.01 (very Good)

Technical skills

Backend Skills :

Java , Object-Oriented Programming (OOP) , Java Collections Framework , Spring Boot , RESTful APIs
Dependency Injection (DI) , Maven / Gradle , Authentication & Authorization (JWT, OAuth) , Hibernate / JPA ,
Exception Handling , API Documentation

Frontend Skills :

HTML5 , CSS3 , JavaScript (ES6+) , Responsive Web Design , Flexbox & CSS Grid , DOM
Manipulation , Fetch API & AJAX , JSON , React.js , State Management , Browser Developer Tools .

Database Skills :

SQL , MySQL , MongoDB , Database Design , ERD (Entity Relationship Diagrams) , Normalization , Query
Optimization , Transactions , Indexing .

Computer Science Fundamentals :

Data Structures , Algorithms , Design Patterns , Problem Solving , Object-Oriented Design , Time & Space
Complexity Analysis , Operating Systems Fundamentals , Computer Networks Fundamentals , Software Engineering
Principles , SOLID Principles

Tools & Technologies :

Git , GitHub , Postman , VS Code , IntelliJ IDEA , Linux Basics

Projects

- **Personal Portfolio Website:** <https://github.com/mohammedi291274smail/som3a-portfolio-website>

Description:

Designed and developed a fully responsive personal portfolio website to showcase my skills, projects, and contact information. The website provides an interactive user experience with modern UI/UX principles and allows visitors to explore my work and get in touch easily.

Project phases:-

- Planning and designing the website structure and user interface.
- Developing responsive web pages using HTML5, CSS3, and JavaScript.
- Implementing modern layouts using Flexbox and CSS Grid.
- Creating interactive UI components and smooth animations.
- Optimizing the website for different screen sizes and devices.
- Integrating a contact form for user communication.
- Applying clean code practices and reusable components.
- Testing cross-browser compatibility and responsiveness.
- Deploying the website using Git and hosting services.

Technologies Used:

- HTML5
- CSS3
- JavaScript (ES6+)
- Responsive Web Design
- Flexbox & CSS Grid

- Git & GitHub

- **Premier Zone Backend System:**

<https://github.com/mohammedi291274smai/Premier-League-Stats-API>

Description :

Developed a Spring Boot backend application for managing and searching English Premier League players' statistics. The system imports player data from CSV files into a PostgreSQL database and exposes RESTful APIs for querying and filtering player information.

Project phases :

- Designed the database schema and JPA entities.
- Implemented a layered architecture (Controller, Service, Repository).
- Developed REST APIs for searching and filtering players.
- Built a CSV data processing module to import player data into PostgreSQL.
- Tested and validated API endpoints using Postman.

Technologies Used :

- Java
- Spring Boot
- Spring Data JPA
- PostgreSQL
- Maven
- REST APIs
- Git & GitHub

- **Baby Photoshop Image Editor:**

https://github.com/mohammedi291274smai/filter-editor_AMY.cpp/blob/main/Filter-Editor_AMY.cpp

Description:

Developed a console-based image processing application in C++ that allows users to apply various editing filters and transformations on images. The project focused on implementing image manipulation algorithms and handling image data at the pixel level.

Project phases:

- Implemented multiple image processing filters including Grayscale, Black & White, Invert Colors, Blur, Edge Detection, Crop, Resize, Rotate, Flip, and Image Merging.
- Applied image transformation and enhancement algorithms using pixel-level manipulation techniques.
- Developed image resizing functionality using Bilinear Interpolation.
- Designed a menu-driven interface for loading, editing, and saving images.
- Supported multiple image formats and image storage operations.

Technologies Used:

- C++
- Object-Oriented Programming (OOP)
- Image Processing
- Data Structures
- Algorithms

Languages

- Very good in English